

Multi-Source Equity Alpha Research

Research Area: Quantitative Research & Systematic Equity

Duration: 8 Weeks

Project Overview

Systematic equity research seeks to identify persistent predictive signals from firm-level and market information that can explain future cross-sectional stock returns. In practice, useful information is distributed across multiple sources, including financial statements, earnings announcements, historical market data, and unstructured financial news.

This project develops a multi-source equity alpha research framework that combines fundamental, earnings, market, and news-based information to construct predictive firm-level signals. The framework will transform raw financial information into standardized features, evaluate individual alpha signals, combine them using statistical and machine-learning models, and translate model predictions into systematic long-short equity portfolios.

The central research question is:

Can information from fundamentals, earnings, market dynamics, and news sentiment be combined to generate persistent out-of-sample cross-sectional equity alpha?

The project emphasizes realistic quantitative research practices, including point-in-time data construction, prevention of look-ahead bias, out-of-sample validation, transaction-cost-aware backtesting, and robustness analysis across market regimes.

Project Objectives

1. Build a point-in-time multi-source equity dataset integrating financial statements, earnings information, market data, and financial news.
2. Construct firm-level alpha features covering valuation, profitability, growth, earnings surprises, momentum, volatility, liquidity, and news sentiment.
3. Evaluate individual predictive signals using cross-sectional Information Coefficients (IC), rank correlations, and portfolio spreads.
4. Develop statistical and machine-learning models to combine multiple weak signals into a unified stock-level alpha score.
5. Perform strict out-of-sample and walk-forward validation to measure whether predictive relationships persist over time.
6. Translate predicted alpha scores into systematic long-short equity portfolios and evaluate returns after turnover and transaction costs.
7. Conduct robustness, factor-exposure, and regime analysis to determine whether observed performance represents distinct and persistent alpha.

Project Timeline & Core Modules

Week	Core Module	Key Tasks	Deliverables
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Week 1	Research Design & Data Architecture	1. Research Universe Definition: Define the initial equity universe using liquid U.S. large-cap stocks, with the S&P 500 as the baseline universe. 2. Prediction Target Design: Define a forward 20-trading-day stock return or cross-sectional return rank as the primary prediction target. 3. Rebalancing Framework: Establish weekly signal generation and portfolio rebalancing rules. 4. Data Source Design: Identify sources for market prices and volume, financial statements, earnings information, and financial news. 5. Research Protocol: Specify the historical sample period, timing conventions, and rules for preventing look-ahead bias.	1. Research specification document 2. Data-source map and data dictionary 3. Initial raw-data collection scripts 4. Prediction and rebalancing framework
Week 2	Multi-Source Data Processing & Alignment	1. Data Cleaning: Handle missing observations, duplicate records, extreme values, and inconsistent identifiers across datasets. 2. Point-in-Time Alignment: Align quarterly financial statements, earnings announcement dates, daily market data, and news timestamps while ensuring that only information available at each historical date is used. 3. Cross-Sectional Preprocessing: Apply winsorization, normalization, and standardized stock identifiers. 4. Dataset Construction: Merge all data sources into a stock-date panel in which each observation contains the information available for a specific company at a specific time. 5. Dataset Splitting: Construct chronological training, validation, and test periods.	1. Cleaned multi-source panel dataset 2. Point-in-time alignment pipeline 3. Data-quality report 4. Reproducible preprocessing code
Week 3	Fundamental & Market Alpha Factor Construction	1. Fundamental Features: Construct valuation, profitability, quality, leverage, and growth measures, including P/E, P/B, ROE, ROA, margins, revenue growth, EPS growth, and leverage. 2. Earnings Features: Create earnings-related variables such as EPS surprise, revenue surprise, earnings growth, and post-earnings price behavior. 3. Market Features: Construct momentum, short-term reversal, realized volatility, trading-volume, liquidity, and beta-related signals. 4. Cross-Sectional Standardization: Winsorize extreme observations and convert raw factors into standardized cross-sectional z-scores or percentile ranks. 5. Individual Signal Testing: Evaluate predictive power using Information Coefficient, Rank IC, quantile returns, and top-minus-bottom portfolio spreads.	1. Firm-level factor library 2. Factor construction notebook 3. Individual factor tear sheets 4. Baseline IC and portfolio-spread results

Week 4	Financial News NLP & Sentiment Signals	1. News Processing: Clean financial headlines and articles, remove duplicate content, and match news observations to the corresponding publicly traded companies. 2. Sentiment Modeling: Apply a financial-domain NLP model such as FinBERT to estimate positive, neutral, and negative sentiment scores. 3. News Feature Engineering: Construct features including average sentiment, sentiment change, negative-news intensity, abnormal news volume, and recent news frequency. 4. Signal Aggregation: Aggregate article-level predictions into company-level daily and weekly sentiment features. 5. Predictive Analysis: Test whether sentiment signals predict future returns over different holding horizons.	1. Financial-news NLP pipeline 2. Company-level sentiment dataset 3. News alpha-factor library 4. Sentiment predictive-power report
Week 5	Multi-Source Alpha Modeling	1. Baseline Models: Develop linear and regularized regression models to provide interpretable benchmarks for multi-factor return prediction. 2. Machine-Learning Models: Train nonlinear models such as Random Forest, XGBoost, or LightGBM using fundamental, earnings, market, and news features. 3. Alpha Score Construction: Transform model outputs into cross-sectional stock rankings representing expected relative performance. 4. Feature Interaction Analysis: Investigate nonlinear relationships among valuation, momentum, earnings, and sentiment variables. 5. Model Interpretability: Use feature importance and SHAP analysis to identify the variables driving model predictions.	1. Baseline statistical models 2. Multi-source ML alpha models 3. Cross-sectional alpha scores 4. Feature-importance and SHAP analysis
Week 6	Out-of-Sample Validation & Signal Selection	1. Walk-Forward Validation: Implement rolling or expanding-window training procedures so that all model predictions are generated using only historically available information. 2. Model Comparison: Compare individual factors, linear multi-factor models, and machine-learning models using out-of-sample performance. 3. Quantitative Evaluation: Measure Information Coefficient, Rank IC, IC Information Ratio, hit rate, and top-minus-bottom return spreads. 4. Feature Ablation: Remove fundamental, earnings, market, and news feature groups individually to measure their incremental predictive contribution. 5. Final Signal Selection: Identify the most stable model specification for portfolio construction.	1. Out-of-sample validation framework 2. Model comparison report 3. Signal-ablation analysis 4. Selected final alpha model

Week 7	Portfolio Construction & Long-Short Backtesting	1. Portfolio Formation: Rank stocks by predicted alpha and construct long positions in the highest ranked stocks and short positions in the lowest ranked stocks. 2. Weighting Design: Compare equal-weighted and signal-weighted portfolio construction. 3. Risk Controls: Introduce position limits, sector constraints, and optional market or beta neutralization. 4. Transaction-Cost Modeling: Incorporate portfolio turnover, trading costs, and execution assumptions. 5. Performance Evaluation: Calculate annualized return, volatility, Sharpe ratio, maximum drawdown, turnover, and long-short equity curves.	1. Long-short backtesting engine 2. Portfolio equity curves 3. Performance-metric report 4. Transaction-cost analysis 5. Portfolio-risk diagnostics
Week 8	Robustness Analysis & Final Research Delivery	1. Market-Regime Analysis: Evaluate performance across bull, bear, high-volatility, and low-volatility market environments. 2. Subperiod & Sector Analysis: Test whether predictive performance persists across different time periods and industry sectors. 3. Parameter Robustness: Compare alternative prediction horizons, rebalance frequencies, portfolio sizes, and transaction-cost assumptions. 4. Factor Exposure Analysis: Measure exposure to established style factors such as market, size, value, momentum, and volatility to determine whether the model captures distinct alpha. 5. Final Research Compilation: Consolidate methodology, experiments, visualizations, results, limitations, and conclusions into a complete research package.	1. Robustness-analysis report 2. Factor-exposure analysis 3. Final research notebook / GitHub repository 4. Final research report 5. Presentation deck

Key Project Milestones

- **Milestone 1 (End of Week 2):** Complete the point-in-time multi-source data pipeline and produce a clean stock-date research panel integrating fundamental, earnings, market, and news information.
- **Milestone 2 (End of Week 4):** Complete the factor and NLP feature libraries and identify the predictive characteristics of individual fundamental, market, earnings, and news signals.
- **Milestone 3 (End of Week 6):** Complete out-of-sample model comparison and determine the final multi-source alpha modeling approach.
- **Milestone 4 (End of Week 8):** Deliver the complete systematic equity research framework, including signal generation, machine-learning modeling, long-short portfolio backtesting, transaction-cost analysis, and robustness testing.

Required Tools

- **Programming:** Python, Jupyter Notebook, Git, GitHub.
- **Data Processing:** Pandas, NumPy, Polars or similar tabular-data tools.
- **Statistical & Machine Learning:** Scikit-learn, XGBoost, LightGBM, SHAP.
- **Natural Language Processing:** PyTorch, Hugging Face Transformers, FinBERT or related financial-language models.
- **Market & Fundamental Data:** Historical price and volume data, company financial statements, earnings-announcement data, and financial-news datasets.
- **Backtesting & Analytics:** Custom Python backtesting framework with support for cross-sectional ranking, portfolio turnover, transaction costs, risk exposure, and performance attribution.

Core Evaluation Metrics

The predictive component of the framework will emphasize cross-sectional ranking rather than only point-estimation accuracy. For each date t , the Information Coefficient may be defined as

$$IC_t = \text{Corr}(\hat{r}_{i,t}, r_{i,t \rightarrow t+20}),$$

where $\hat{r}_{i,t}$ represents the predicted alpha score and $r_{i,t \rightarrow t+20}$ represents the subsequent 20-trading-day return.

Model evaluation will include:

- Information Coefficient (IC)
- Rank Information Coefficient
- IC Information Ratio
- Top-minus-bottom portfolio return
- Annualized return
- Annualized volatility
- Sharpe ratio
- Maximum drawdown
- Portfolio turnover
- Transaction-cost-adjusted performance

Reference Papers

Fama, E. F., & French, K. R. (1993). *Common Risk Factors in the Returns on Stocks and Bonds*.

Jegadeesh, N., & Titman, S. (1993). *Returns to Buying Winners and Selling Losers: Implications for Stock Market Efficiency*.

Fama, E. F., & French, K. R. (2015). *A Five-Factor Asset Pricing Model*.

Gu, S., Kelly, B., & Xiu, D. (2020). *Empirical Asset Pricing via Machine Learning*.

Kelly, B., Pruitt, S., & Su, Y. (2019). *Characteristics Are Covariances: A Unified Model of Risk and Return*.

Ke, Z. T., Kelly, B., & Xiu, D. (2019). *Predicting Returns With Text Data*.